Sinusoids and Phasor Analysis

Video 12

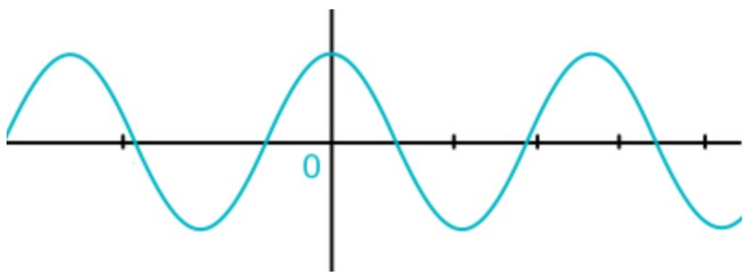
Electrical Science

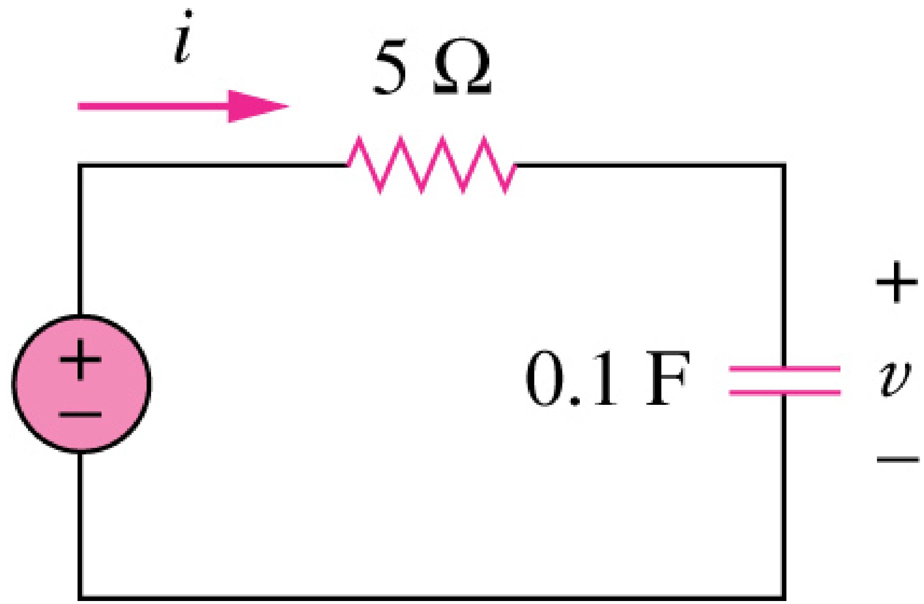
Dr. Mohamed Bingabr
University of Central Oklahoma

Sinusoids and Phasor

- Sinusoids' features
- Phasors
- Phasor relationships for circuit elements
- Impedance and admittance
- Kirchhoff's laws in the frequency domain
- Impedance combinations

How to determine *v(t)* and *i(t)*?





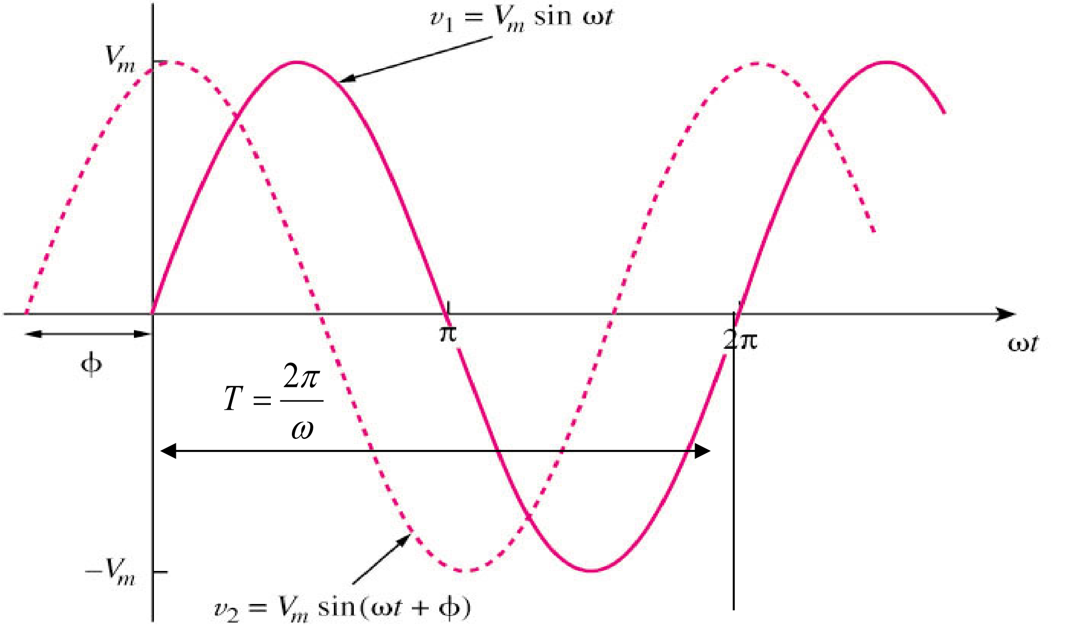
How can we apply what we have learned before to determine *i(t)* and *v(t)*?

Sinusoids

- A sinusoid is a signal that has the form of the sine or cosine function.
- A general expression for the sinusoid,

$$v(t) = V_m \sin(\omega t + \phi)$$

$$f = \frac{1}{T} Hz \quad \omega = 2\pi f$$



where

V_m = the **amplitude** of the sinusoid
ω = the angular frequency in radians/s
Φ = the phase

A periodic function is one that satisfies *v(t)* = *v(t + nT)*, for all t and for all integers n.

Sinusoids

Example 1

Given a sinusoid, 5 sin(4πt - 60°), calculate its amplitude, phase, angular frequency, period, and frequency.

Solution:

Amplitude = 5, phase = –60°, angular frequency = 4π rad/s, Period = 0.5 s, frequency = 2 Hz.

Sinusoids

Example 2

Find the phase angle between i_1 = -4 sin(377t + 25^o), and i_2 = 5 cos(377t - 40^o), does i_1 lead or lag i_2?

Solution:

Since sin(ωt+90^o) = cos ωt

$$i_2 = 5\sin(377t - 40^o + 90^o) = 5\sin(377t + 50^o)$$

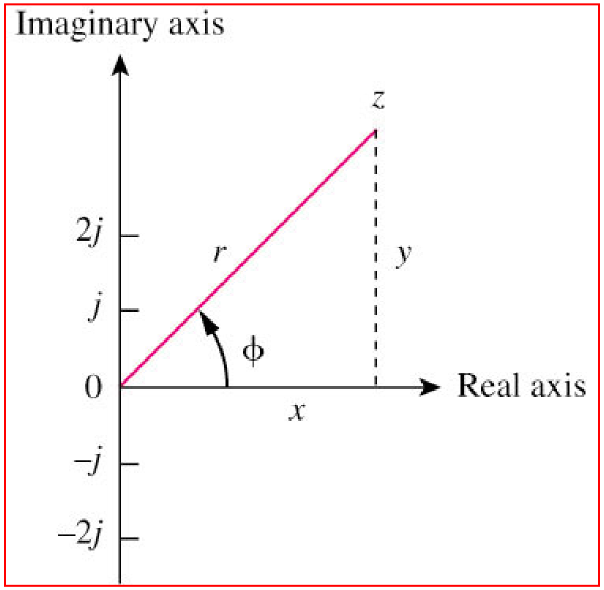
$$i_1 = -4\sin(377t + 25^o) = 4\sin(377t + 180^o + 25^o) = 4\sin(377t + 205^o)$$

therefore, i_1 leads i_2 155^o.

Phasor

- A phasor is a complex number that represents the amplitude and phase of a sinusoid.

- It can be represented in one of the following three forms:



a. Rectangular $z = x + jy = r(\cos\phi + j\sin\phi)$

b. Polar $z = r\angle\phi$

c. Exponential $z = re^{j\phi}$

where $r = \sqrt{x^2 + y^2}$

$\phi = \tan^{-1}\frac{y}{x}$

Phasor

Example 3

- Evaluate the following complex numbers:

a. $$[(5+j2)(-1+j4)-5\angle 60^{\circ}]$$

b. $$\frac{10+j5+3\angle 40^{\circ}}{-3+j4}+10\angle 30^{\circ}$$

Solution:

a. −15.5 + j13.67

b. 8.293 + j2.2

Phasor

Mathematic operation of complex number:

1. Addition $$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$
2. Subtraction $$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$
3. Multiplication $$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$$
4. Division $$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$$
5. Reciprocal $$\frac{1}{z} = \frac{1}{r} \angle -\phi$$
6. Square root $$\sqrt{z} = \sqrt{r} \angle \phi/2$$
7. Complex conjugate $$z^* = x - jy = r \angle -\phi = re^{-j\phi}$$
8. Euler's identity $$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

Cosine Function in Phasor

- Transform a sinusoid to and from the time domain to the phasor domain:

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow V = V_m \angle \phi$$

(time domain) (phasor domain)

- Amplitude and phase difference are two principal concerns in the study of voltage and current sinusoids.
- Phasor will be defined from the cosine function in all our proceeding study. If a voltage or current expression is in the form of a sine, it will be changed to a cosine by subtracting from the phase.

Phasor

Example 4

Transform the following sinusoids to phasors:

$i = 6\cos(50t - 40^o)$ A

$v = -4\sin(30t + 50^o)$ V

Solution:

a. $I = 6\angle -40^\circ$ A

b. Since $-\sin(A) = \cos(A+90^o)$;

$v(t) = 4\cos(30t+50^o+90^o) = 4\cos(30t+140^o)$ V

Transform to phasor => $V = 4\angle 140^\circ$ V

Phasor

Example 5:

Transform the following phasors to sinusoids:

a. $\mathbf{V} = -10\angle 30^\circ \text{ V}$

b. $\mathbf{I} = j(5 - j12) \text{ A}$

Solution:

a) $v(t) = 10\cos(\omega t + 210^o)$ V

b) Since $\mathbf{I} = 12 + j5 = \sqrt{12^2 + 5^2}\angle \tan^{-1}(\frac{5}{12}) = 13\angle 22.62^\circ$

$i(t) = 13\cos(\omega t + 22.62^o)$ A

Differences between v(t) and V

- v(t) is instantaneous or time-domain representation
  V is the frequency or phasor-domain representation.
- v(t) is time dependent, V is not.
- v(t) is always real with no complex term, V is generally complex.

Note: Phasor analysis applies only when frequency is constant; when it is applied to two or more sinusoid signals only if they have the same frequency.

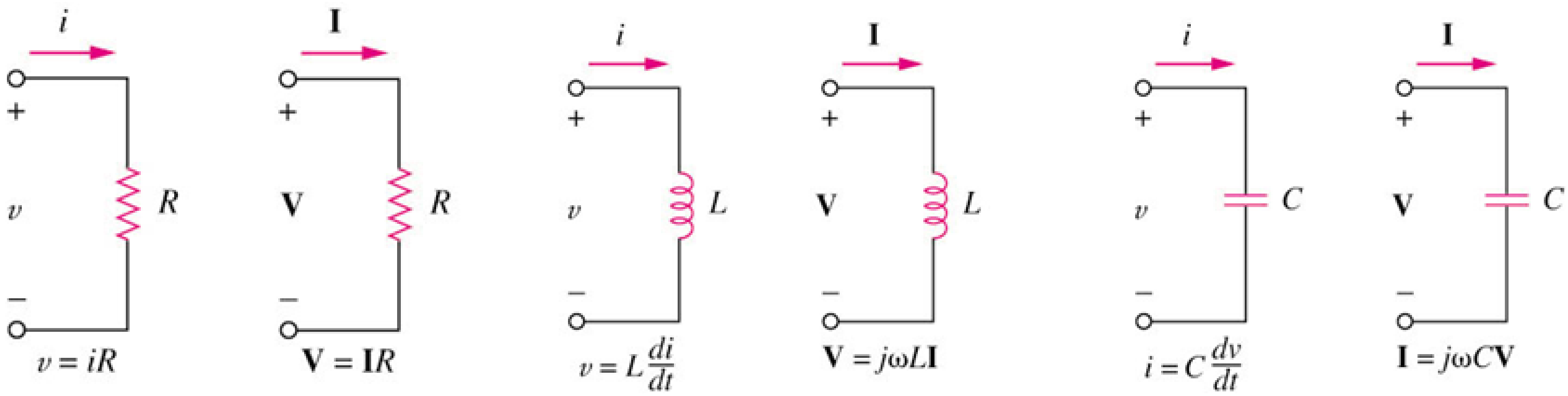
$$v(t) \longleftrightarrow V = V\angle\phi$$

$$\frac{dv}{dt} \longleftrightarrow j\omega V$$

$$\int v\,dt \longleftrightarrow \frac{V}{j\omega}$$

Phasor and Impedance of Circuit

Time Phasor Time Phasor Time Phasor



Z = R Z = $j\omega L$ Z = $1/j\omega C$

- The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I, measured in ohms Ω.

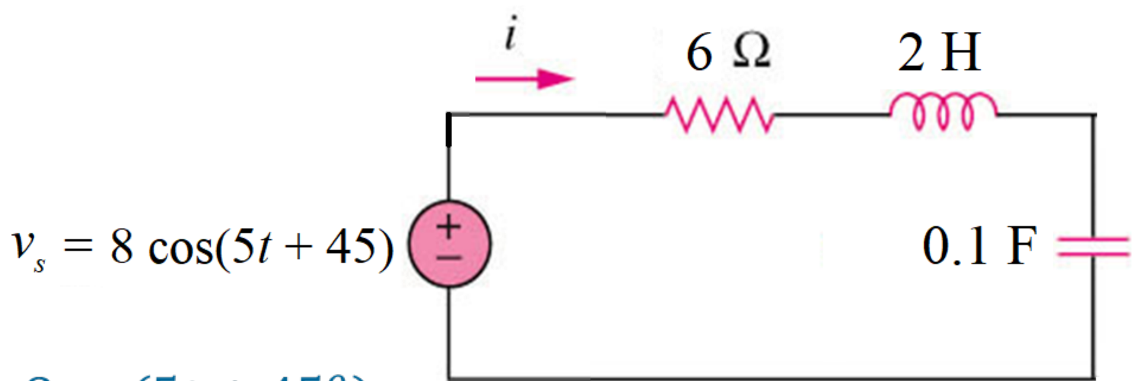
$$Z = \frac{V}{I} = R + jX$$

where R = Re[Z] is the resistance and X = Im[Z] is the reactance. Positive X is for L and negative X is for C.

Impedance of inductor is zero (short circuit) for DC and infinity (open circuit) for high frequency.
Impedance of capacitor is zero (short circuit) for high frequency and infinity (open circuit) for DC.

Phasor/Frequency Analysis of Circuit

- To find the output i(t) in the time domain we find the differential equation that relate i(t) to the input then solve it.



$$6i(t) + 2\frac{di(t)}{dt} + \frac{1}{0.1}\int i(t)dt = 8\cos(5t + 45^o)$$

- However, the derivation may sometimes be very tedious.

Instead of first deriving the differential equation and then transforming it into phasor to solve for I, we can transform all the RLC components into phasor first, then apply the KCL laws and other theorems to set up a phasor equation involving I directly.

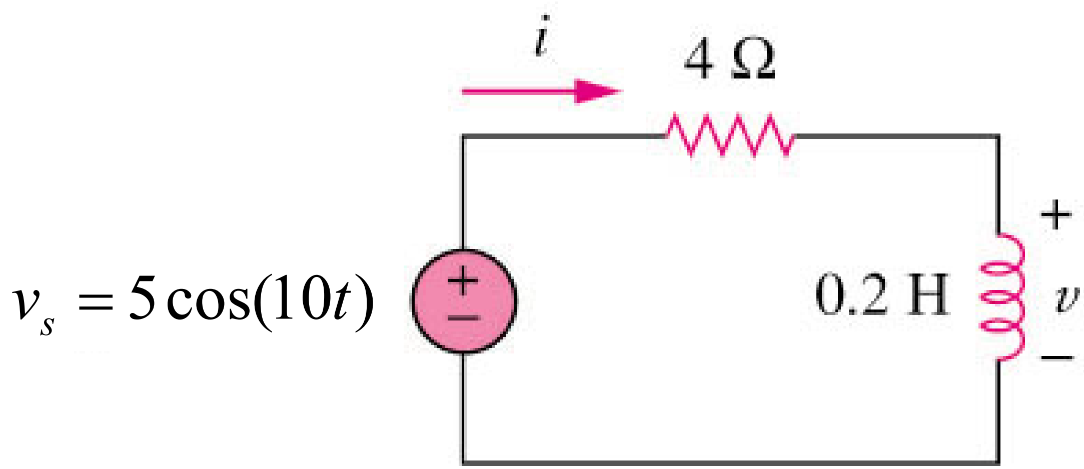
Kirchhoff's Laws in the Frequency Domain

- Both KVL and KCL are hold in the phasor domain or more commonly called frequency domain.
- The following principles used for DC circuit analysis all apply to AC circuit such as voltage division, current division, circuit reduction, impedance equivalence, and Y-Δ transformation.
- Moreover, the variables to be handled are phasors, which are complex numbers.
- All the mathematical operations involved are now in complex domain.

Impedance and Admittance

Example 6

Refer to Figure below, determine *v(t)* and *i(t).*

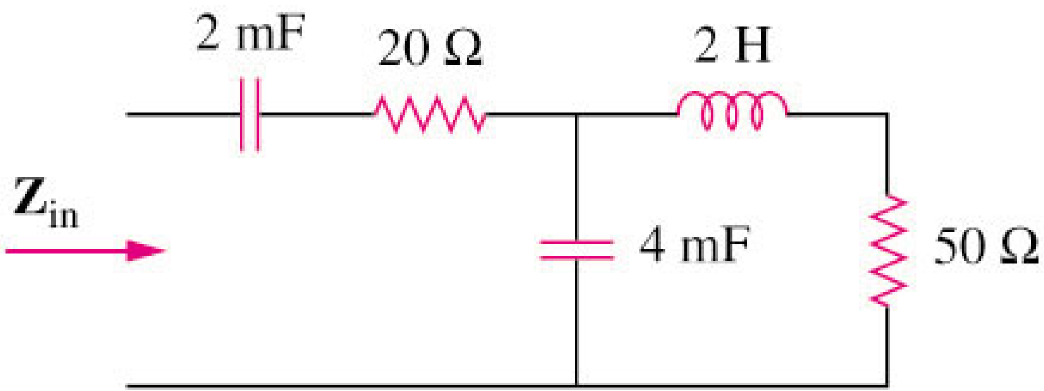


Answers: *i(t) = 1.118cos(10t – 26.56°) A; v(t) = 2.236cos(10t + 63.43°) V*

Impedance Combinations

Example 7

Determine the input impedance of the circuit in figure below at $\omega = 10$ rad/s.



Answer: Z_{in} = **32.38 – j73.76**